33. THE FEMALE GENITAL SYSTEM

DURATION : 06:15 | STUDY SHEET

COURSE SUMMARY

In this video, we delve deeply into the development of the female reproductive system, starting with the formation of the Wolffian and Müllerian ducts, and their impact on reproductive structure. The focus is on the uro-genital relationship, which differs between men and women, highlighting the creation of the uterus and fallopian tubes. We also explore the role of the broad and ovarian ligaments, as well as the importance of kidney health in hormonal regulation and fertility. The video addresses fascial continuity in the pelvis and how anomalies such as fibroids can influence overall body dynamics, providing crucial information for osteopathic practitioners and those interested in embryonic development.

THE FEMALE GENITAL SYSTEM

1. MESONEPHRIC DUCT (WOLFFIAN DUCT) AND PARAMESONEPHRIC DUCT (MÜLLERIAN DUCT)

The **Wolffian duct**, also known as the **mesonephric duct**, is the first duct to appear in the embryo.

In males, this duct descends and is integrated between the testes to form the **ejaculatory duct**. In contrast, the Müllerian duct remains as a vestige and contributes to the formation of the core of the prostate.

In females, the situation is reversed:

- The Wolffian duct differentiates into **Gardner's organs**, which are vestiges.
- The Müllerian duct, through fusion, becomes the **uterus** and **fallopian tubes**.

UROGENITAL DEVELOPMENT

The **urogenital** relationship is fundamental. In males, the epididymal duct, vas deferens, and ejaculatory duct form from this relationship.

In females, this development does not occur in the same way. The Müllerian duct meets at the midline to form the uterus.

Gardner's organs represent the vestiges of the Wolffian duct in females. The ejaculatory duct is oriented towards the base, and the further away it is, the more it will form the uterus. The closer it is, the more the vestige will be the **prostatic utricle**.

2. THE BROAD LIGAMENT AND OVARIAN LIGAMENTS

In both males and females, the **posterior parietal peritoneum** concentrates to form the **broad ligament**, as well as the small ovarian ligaments and round ligaments.

It is the same tissue, but with different concentrations. This posterior parietal peritoneum will form the uterus in females.

In males, this movement does not exist, and only a concentration of peritoneum remains.

3. THE KIDNEY AND ADRENAL GLAND

The kidney is located below the **adrenal gland**, separated by a capsule. This means that a **renal ptosis** does not lead to adrenal ptosis.

They are separated by a fascia called **Glisson's capsule**. The kidney requires a fatty space.

In very thin women, kidney ptosis can lead to menstrual irregularities and reduced fertility. **Revitalising a kidney** also means revitalising its fertility.

EMBRYONIC DEVELOPMENT OF THE KIDNEY

Initially, in a small embryo, the kidneys are enormous, as is the liver. This is reminiscent of brain development, which is very developed compared to the trunk at first.

These proportions evolve during development. The liver plays an important role in the development of the adrenal glands.

This is the first time that a **structural differentiation** between male and female is observed, linked to the greater development of the glands that create the **Müllerian tubercle**.

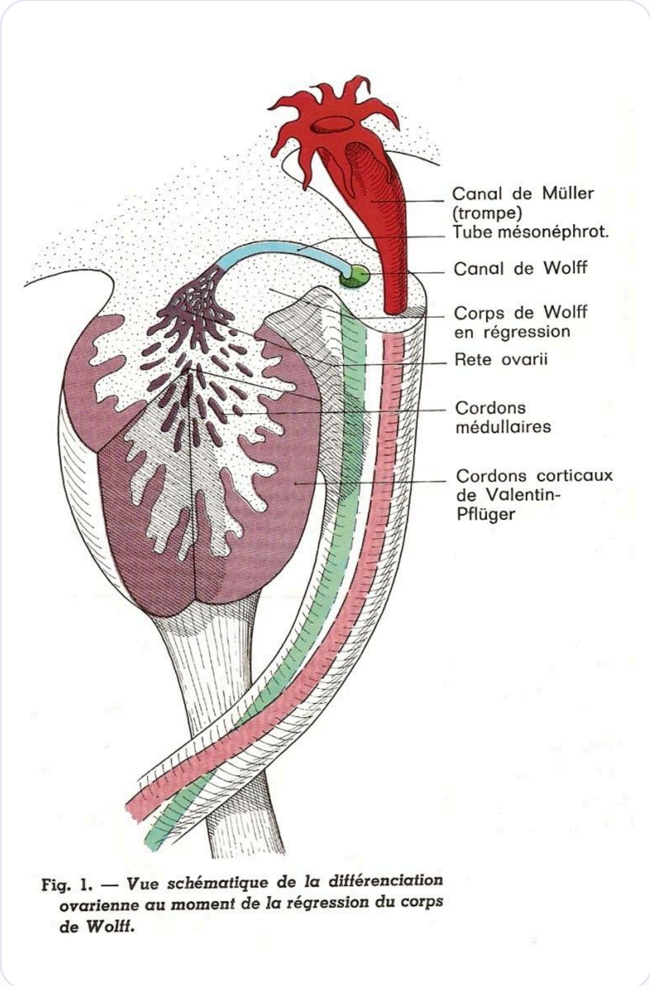

FIG. — Ovarian differentiation

4. DEVELOPMENT OF THE MÜLLERIAN SYSTEM (5TH WEEK)

By the fifth week, the system on the side begins to develop, restoring full power to the kidney.

The brain influences the establishment of the heart, which in turn establishes **diaphragmatisation**, the liver, adrenal glands, gonads, and the ridge.

The Müllerian duct develops much more strongly in females to create a pure canal, which joins to form the uterus.

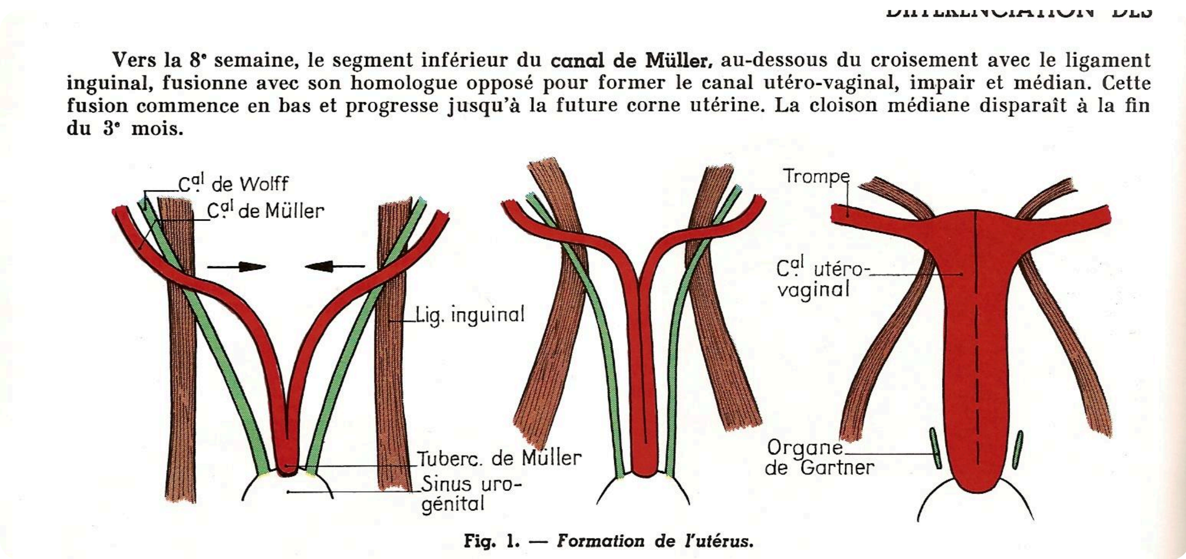
Vers la 8e semaine, le segment inférieur du **canal de Müller**, au-dessous du croisement avec le ligament inguinal, fusionne avec son homologue opposé pour former le canal utéro-vaginal, impair et médian. Cette fusion commence en bas et progresse jusqu'à la future corne utérine. La cloison médiane disparaît à la fin du 3e mois.

Fig. 1. — *Formation de l'utérus.*

FIG. — Formation of embryonic uterus

In males, development is less pronounced, and what remains at the bottom of the Müllerian duct constitutes the core of the prostate.

TESTICULAR DESCENT

Testicular descent is due to the differentiated growth rate at the **gubernaculum**, which slows down and pulls the entire system.

This fascial continuity between the uterus and the entire part of the fascia, skin, and small pelvis is essential.

5. FASCIAL CONTINUITY AND INFLUENCE OF STRUCTURES

The uterus, ligaments, ovaries, and **peritoneal cavity** are clearly visible.

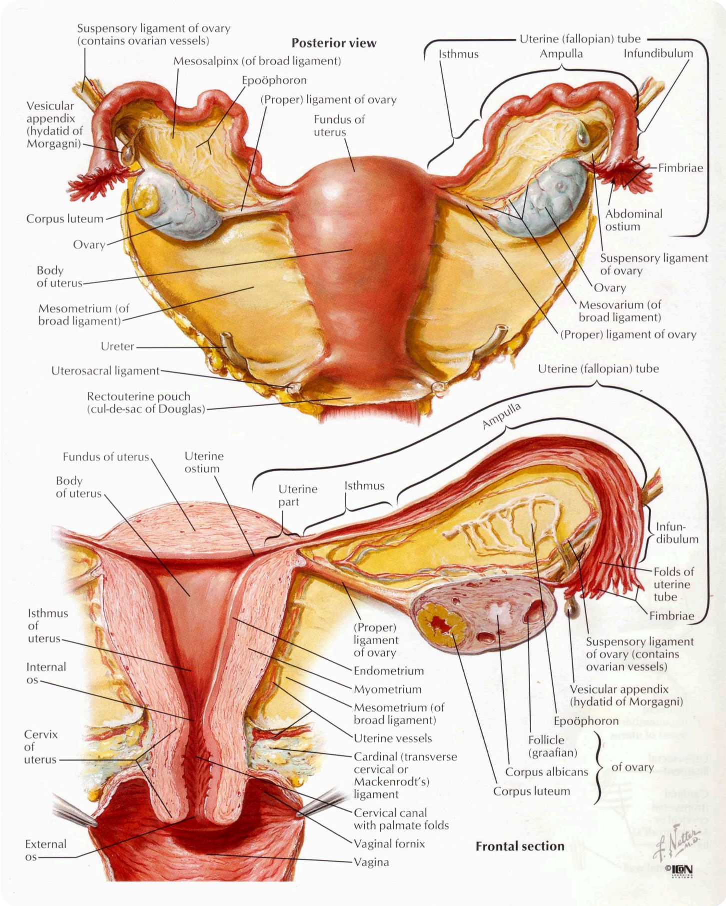

FIG. — Female reproductive system

By lifting the uterus with forceps, one can see that everything moves.

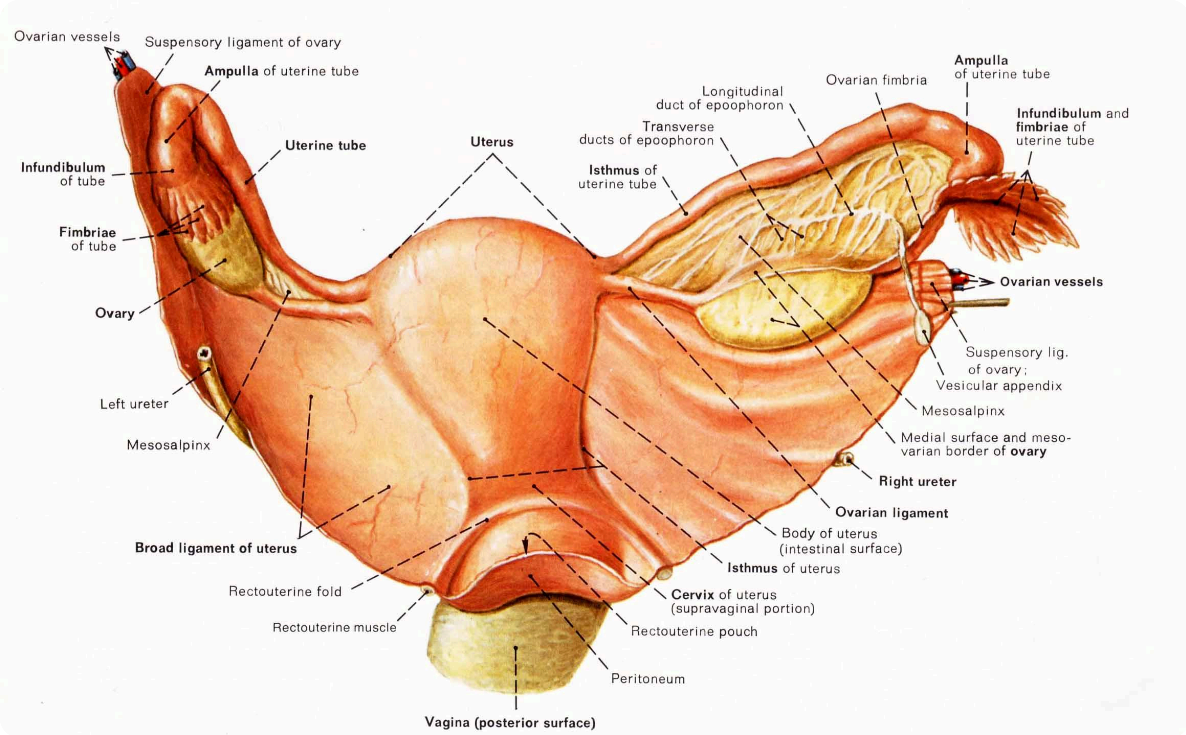

FIG. — Female reproductive system

The impact of a **fibroid** or a **cyst** is significant. Continuity with the ligaments is paramount.

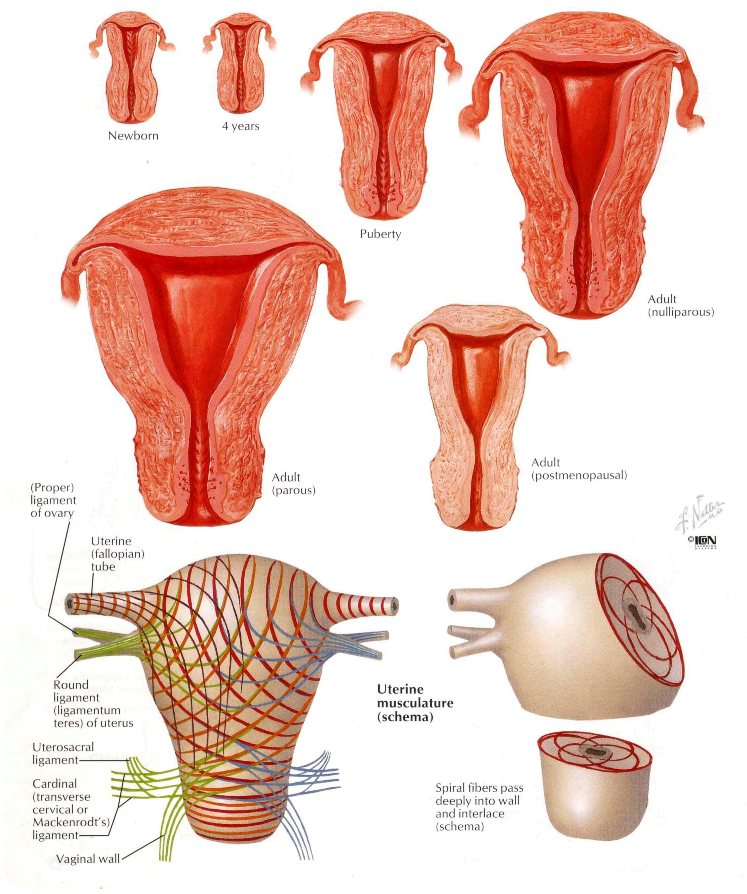

FIG. — Uterus and musculature

When pulled harder, one can see how the entire system reacts, including the round ligaments.

The peritoneum is a very intelligent tissue. The influence of a fibroid, for example, can modify the entire downstream system through a simple traction.

A large fibroid can lead to a cascade of consequences throughout the system.

al de Muller ouvert dans la cavité

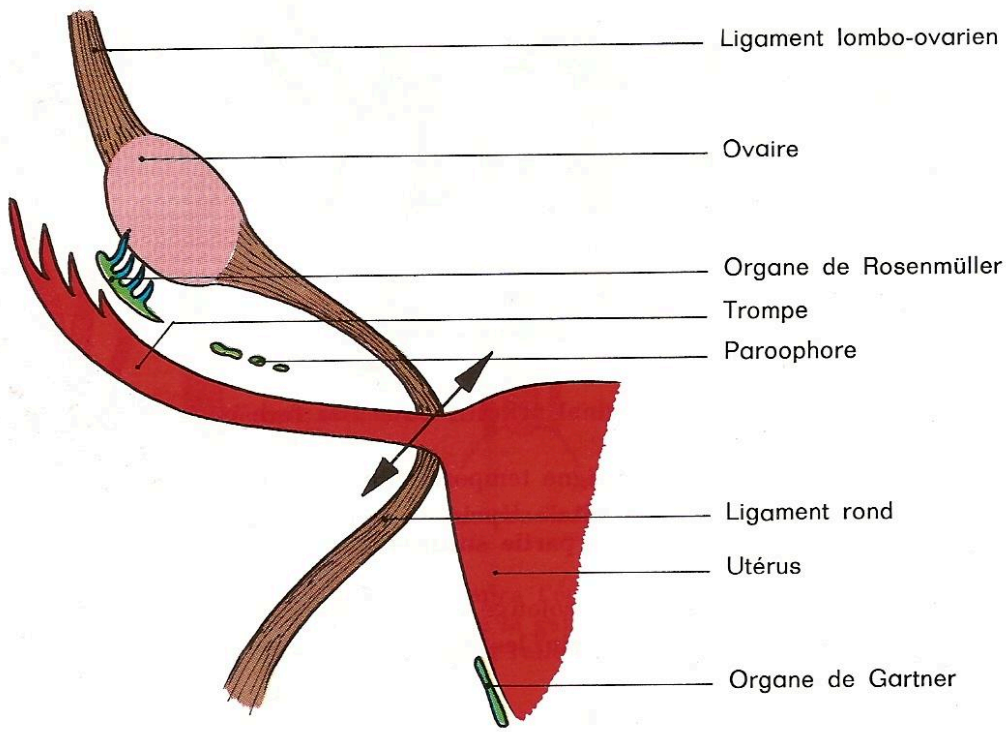

Fig. 1. — *Formation de la trompe et résidus wolffiens.*

FIG. — Female genital organs

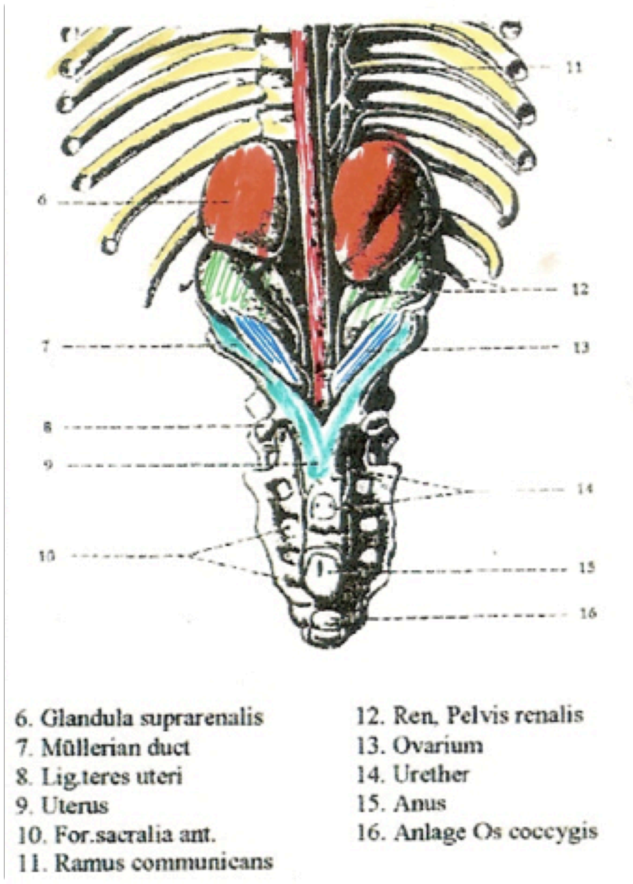

FIG. — Fetal genital system

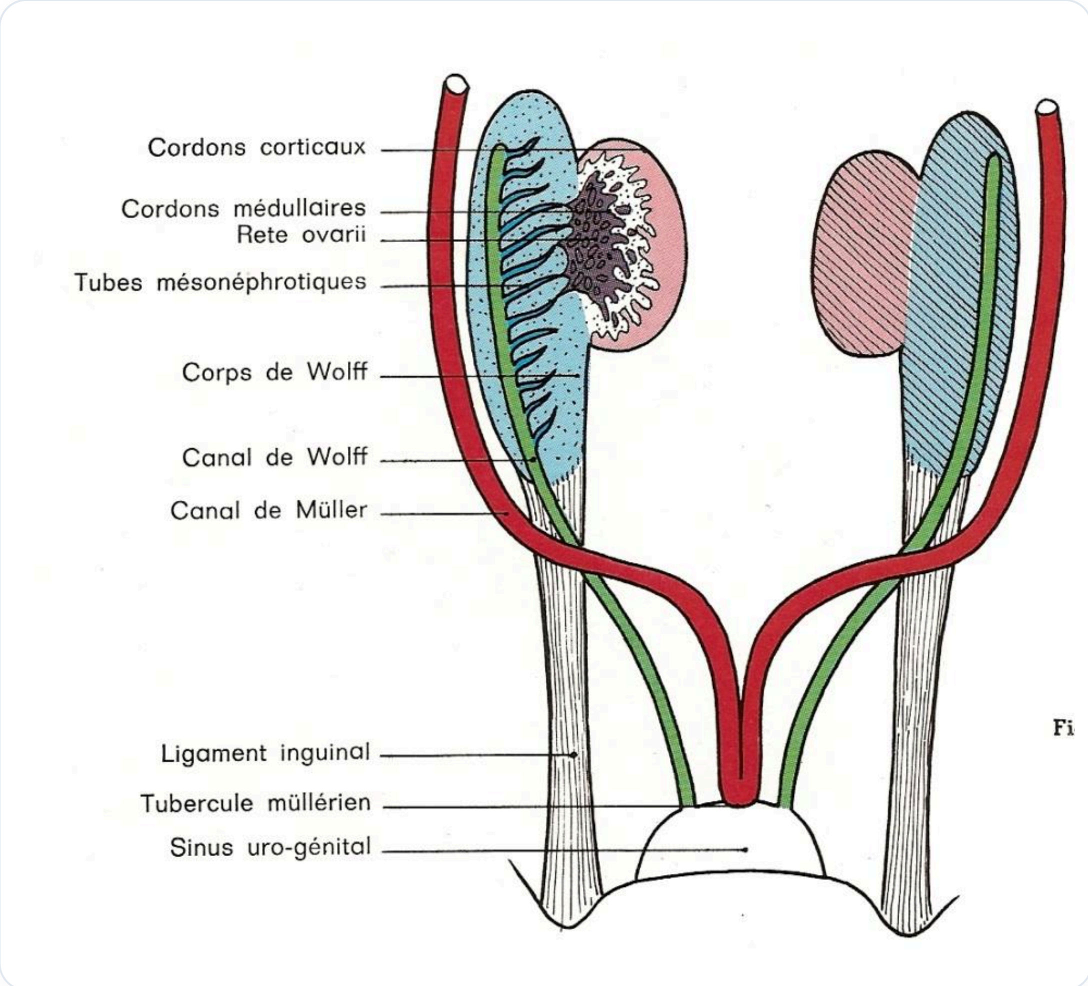

FIG. — Undifferentiated genital system